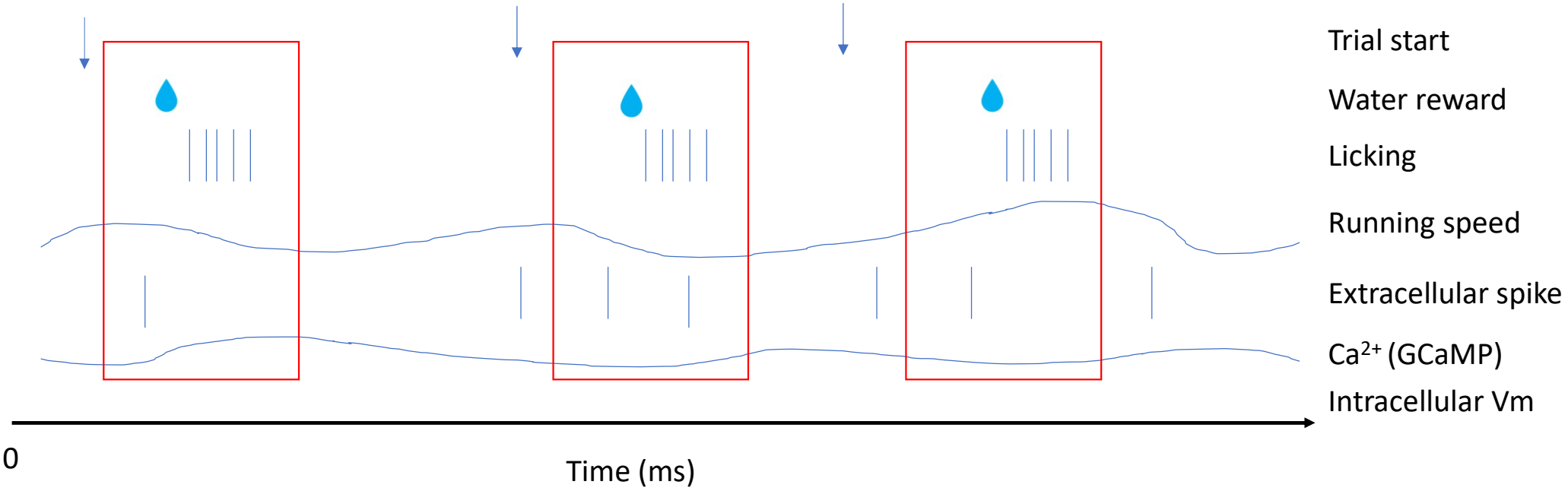
libtimecourse: A Matlab toolbox for neural data analysis

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In a nutshell: what you can do with libtimecourse

Data



- Timestamp signals
  - Task event (e.g., stimulus onset), behavioral events (e.g., licking) or neural activity (e.g., single unit spike)
  - Time in millisecond from session start (0 ms)
- Continuous signals
  - Sampled at 1000Hz (1ms bin) from session start (0 ms)

Motivation

- Provide a set of functions that allow users to manage data flexibly, rigorously, and easy to verify the results.
- This helps the users avoid spending time on
  - Verifying basic properties (e.g., alignment on the trigger) of PSTHs
  - Combining PSTHs and aligning time series across different PSTHs
  - Checking whether some experimental groups are missing
  - Dealing with errors in ‘basic but essential’ data processing parts.
- Users can focus on their own scientific questions
  - E.g., what does the multi-dimensional population activity mean?

Design principle

- Single-session plot
  - Show individual trials and average PSTH
  - Variable time window (e.g., reaction-time tasks)
  - Sort trials by experimental conditions
  - Sort trials by an event of interest
  - Statistical test
  - Save the PSTHs with all the relevent (e.g., group, timing, event) information
- Population plot
  - Manage multiple PSTHs easily
  - Combine different time range
  - Check integrity of experimental conditions across sessions

plot_timecourse(): plotting PSTH

Download and install

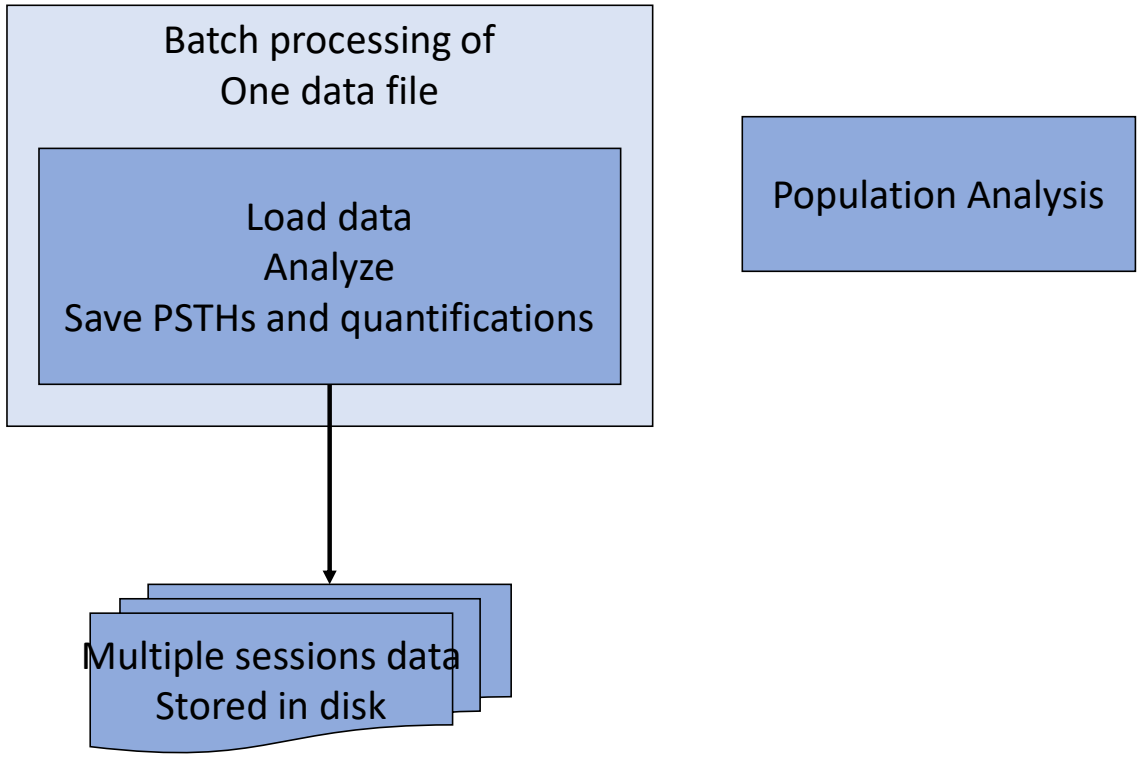
- Go to http://
- Click ‘save’
- Open Matlab

```
>> load libtimecourse_demo_data
>> figure;
>> plot_timecourse(‘timestamp’, ms_lick, ms_reward, -2000, 2000);
```

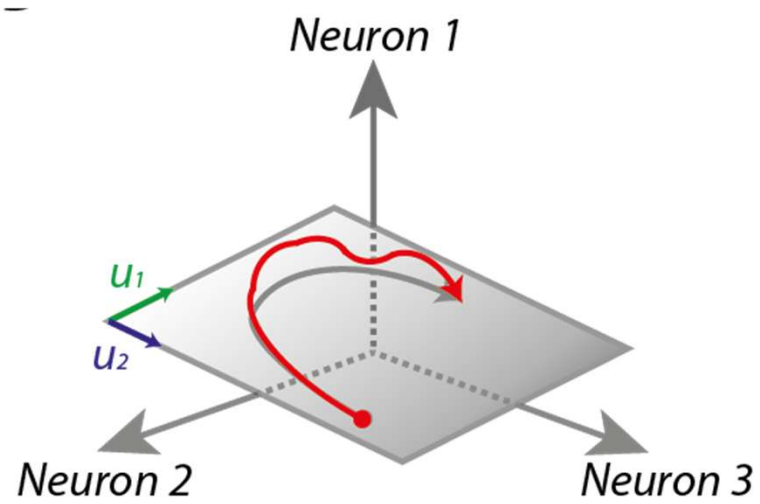
Features

- One function for both timestamp and continuous data
- Time window can be either constant or variable
- Show N for every plot
- Supports simple stats
- easy to save, load, and combine

Data analysis framework

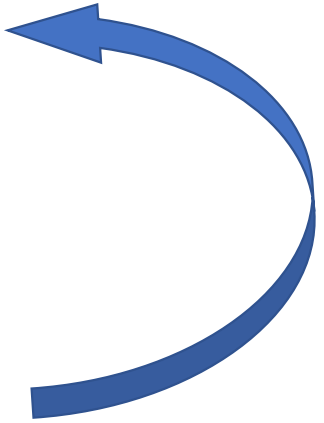


Dynamics in a neural state space



Cognitive process of single-unit analysis

- Hypothesis (or just do assumption-free experiments)
- Repeat until you have N conclusions (e.i., figures)
  - ★Plot results by events of interest
    - Look at the results carefully
    - Make a quantitatively testable hypothesis
  - ★Quantify responses
  - ★Population analysis
  - ★Prove or disprove the hypothesis by statistical tests
    - Do modeling if necessary
- Present
- Write



Fun

IF

- If you can look at data fast, flexibly
- Examine population easily with stats

Otherwise,

- Projects will be delayed
- Conclusion can be wrong

Population analysis

- Extract numbers from PSTHs
- Within-session analysis (across trials)
- Across-session analysis (using session-average, across session)
- Visualization accompanies stats
- Rigorous
  - N, OL